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KOTA (RAJASTHAN)
path to success

# PHYSICS

## JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

## Circular Motion

(English Medium)

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## CIRCULAR MOTION

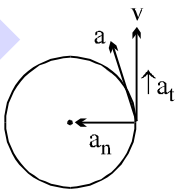
# KEY CONCEPT

**1.** A body moving with constant speed in a circular path is continuously accelerated towards the centre of rotation. The magnitude of this normal acceleration is given by

$$a_n = \frac{v^2}{r} = \omega^2 r$$

where v is the constant speed ($v = \omega r$) and

r is the radius of the circular path




Tangential acceleration : $a_t = \frac{dv}{dt}$, $a = \sqrt{a_t^2 + a_n^2}$

**2.** **Radius of curvature :** $r = \frac{v^2}{a_n}$

**3.** According to Newton's second law, a body moving in a circular path with constant speed must be acted upon by an unbalanced force which is always directed towards the centre. This necessary unbalanced force is called the centripetal force.

$$F = \frac{mv^2}{r} = m\omega^2 r$$

**4.** Centrifugal force is a pseudo force which is observed by an observer in rotating frame.

$$\vec{F}_{cf} = m\omega^2_{frame}\,\vec{r}$$

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# SUPPLEMENT QUESTION

## CROSS PRODUCT SUPPLEMENT:

**1.** $\vec{a}=\hat{i}-2\hat{k}$, $\vec{b}=-\hat{i}+3\hat{j}+\hat{k}$. The value of $\vec{c}=\vec{a}\times\vec{b}$ is :

(A) $6\hat{i}+\hat{j}-2\hat{k}$ (B) $2\hat{i}+\hat{j}$ (C) $2\hat{i}+3\hat{j}+\hat{k}$ (D) $6\hat{i}+\hat{j}+3\hat{k}$

**2.** $\vec{a}=-\hat{i}+2\hat{j}+\hat{k}$ and $\vec{b}=3\hat{i}-2\hat{j}+4\hat{k}$. Find $\vec{b}\times\vec{a}$ and $|\vec{a}\times\vec{b}|$.

**3.** A vector $\vec{A}$ points vertically downward and $\vec{B}$ points towards the north. The direction of $\vec{A}\times\vec{B}$ is

(A) Along west (B) Along east (C) Vertically up (D) Zero

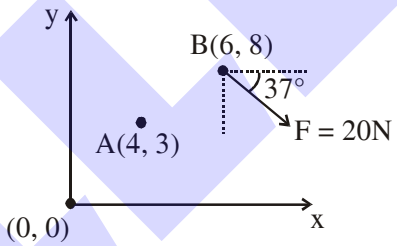
**4.** If $\vec{a}=(2\hat{i}-5\hat{k})\times 3\hat{j}$ and $\vec{b}=\hat{k}\times(-3\hat{j}+4\hat{i})$. Find the value of $\vec{a}+\vec{b}$.

**5.** Evaluate $(\vec{A}+2\vec{B}-\vec{C})\times(3\vec{B}+\vec{C})$. Given that angle between $\vec{A}$ and $\vec{B}$ is 180°, $\vec{A}+5\vec{B}=(2\hat{i}-5\hat{k})$ and $\vec{C}=3\hat{j}$

**6.** Draw the direction of vector $\vec{c}$ if $\vec{c}=\vec{a}\times\vec{b}$

(i) $\vec{b}\odot\longrightarrow\vec{a}$ (ii) $\vec{a}\otimes\longleftarrow\vec{b}$ (iii) $\vec{a}$ $\vec{b}$ (iv) $\vec{a}\uparrow$ $\otimes\vec{b}$

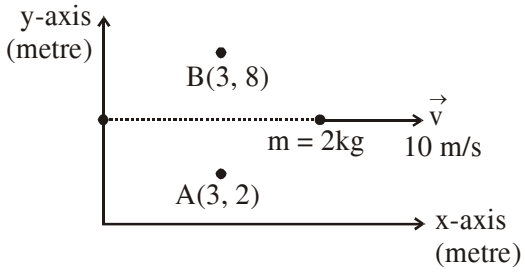
**7.** Torque about a point (say A) is defined as $\vec{\tau}=\vec{r}\times\vec{F}$ where $\vec{F}$ is force and $\vec{r}$ is a vector joining point A to the point of application of force (say B) direction of $\vec{r}$ is from A to B. Find the value of torque about A according to the following figure. (coordinates are in m)




**8.** Angular momentum of a particle about a point (say A) is defined as $\vec{L}=\vec{r}\times\vec{p}$ where $\vec{p}$ is the linear momentum of the particle and $\vec{r}$ is a vector joining point A to use instantaneous position of the particle (say P) with direction from A to P. Taking mass of particle to be m = 2 kg. Assume that the particle is moving on line y = 4. Refer to the figure shown. Find.

(i) $\vec{L}_A$ (Angular momentum about A)

(ii) $\vec{L}_B$ (Angular momentum about B)

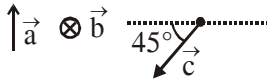


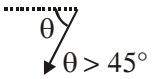

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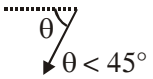
**9.** Draw the direction of $\vec{A}\times(\vec{B}\times\vec{C})$

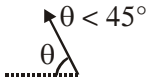
$\vec{A}\odot \longrightarrow \vec{B}$ $\vec{C}\otimes$

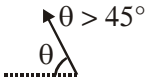
**10.** Draw the direction of $(\vec{A}\times\vec{B})+(\vec{B}\times\vec{C})$ if




(A) 


(B) 


(C) 


(D) 


**11.** If $\vec{A}=2\hat{i}-3\hat{j}+7\hat{k}$, $\vec{B}=\hat{i}+2\hat{j}$ and $\vec{C}=\hat{j}-\hat{k}$.

(i) Find the value of $\vec{A}\cdot(\vec{B}\times\vec{C})$.

(ii) Are the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ coplaner ?

**12.** If two adjacent sides of a parallelogram are given by $\vec{a}=-\hat{i}+2\hat{j}+\hat{k}$ and $\vec{b}=3\hat{i}-2\hat{j}+4\hat{k}$ then the area of parallelogram is :

(A) $\sqrt{165}$ (B) $\sqrt{525}$ (C) $\sqrt{425}$ (D) None of these

**13.** Find a unit vector perpendicular to $\vec{A}$ and $\vec{B}$ both. Also, the vector should have positive component in y-direction. ( $\vec{A}=2\hat{i}+3\hat{j}+\hat{k}$ and $\vec{B}=\hat{i}-\hat{j}+\hat{k}$ )

**14.** If $|\vec{A}\times\vec{B}|=\frac{4}{3}\vec{A}\cdot\vec{B}$ then the value of $|\vec{A}+\vec{B}|$ is:

(A) $(A^2+B^2+AB)^{1/2}$ (B) $(A^2+B^2+\frac{8}{5}AB)^{1/2}$

(C) $A+B$ (D) $(A^2+B^2+\frac{6}{5}AB)^{1/2}$

**15.** If $\vec{A}$ is towards the north, $\vec{B}$ is towards the west and $\vec{C}$ is towards vertically up direction then the direction of $\vec{A}+\vec{C}\times(\vec{A}\times\vec{B})$ is :

(A) North (B) South (C) East (D) West

**16.** A particle moves on a given straight line with a constant speed v. At a certain time it is at a point P on its straight line path. O is a fixed point. Show that $\overrightarrow{OP}\times\vec{v}$ is independent of the position P.

**17.** The force on a charged particle due to electric and magnetic fields is given $\vec{F}=q\vec{E}+q\vec{v}\times\vec{B}$. Suppose $\vec{E}$ is along the X-axis and $\vec{B}$ along the Y-axis. In what direction and with what minimum speed v should a positively charged particle be sent so that the net force on it is zero ?

**18.** A force $\vec{F}=\vec{v}\times\vec{A}$ is exerted on a particle in addition to the force of gravity, where $\vec{v}$ is the velocity of the particle and $\vec{A}$ is a constant vector in the horizontal direction. With what minimum speed a particle of mass m be projected so that it continues to move undeflected with a constant velocity ?

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